Common theories and their applications

Nineteen topics, forty-three original manuscripts

For each topic, $S = \sum_i c(L_i)$ is the length of the complete original manuscripts, U the length of the whole common theory, and $A = \sum_i c(A_i)$ the length of all complete applications. We report

$$R = \frac{S}{U + A}, \qquad r_i = \frac{c(L_i)}{c(A_i)}.$$

The second comparison assumes acquisition of the whole common theory and the declared required dependencies. Each topic pays its common theory independently. A factor greater than one indicates a shorter written replacement.

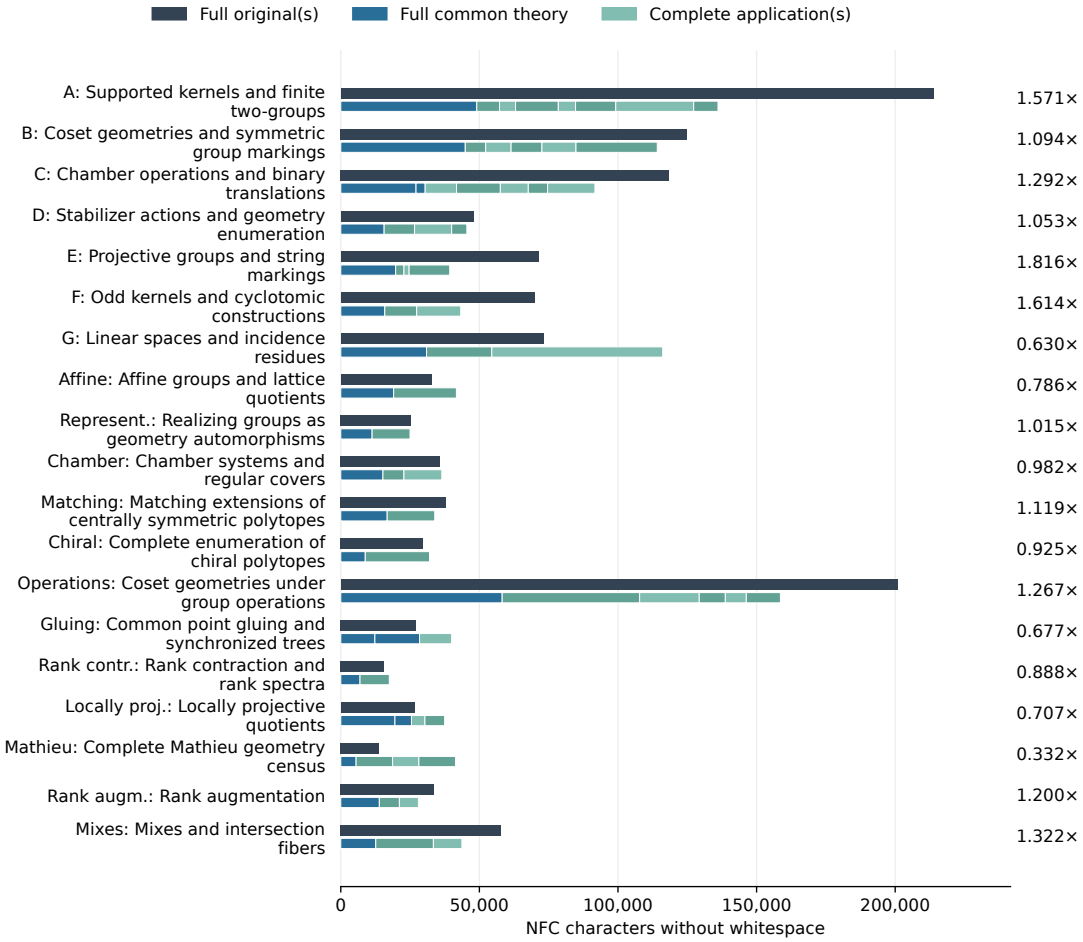
Complete manuscript costs. The unit is NFC characters without whitespace, at each topic's stated representation. Introductions, context, proofs, companions and references remain included.

Topic	Originals	Sources	Common	Applications	Written	Factor
A	7	213,803	49,059	87,019	136,078	1.571
B	5	124,906	44,905	69,265	114,170	1.094
C	5	118,433	30,437	61,248	91,685	1.292
D	3	47,894	15,632	29,855	45,487	1.053
E	3	71,375	19,790	19,513	39,303	1.816
F	2	69,825	15,876	27,398	43,274	1.614
G	2	73,199	30,928	85,207	116,135	0.630
Affine	1	32,767	19,070	22,638	41,708	0.786
Represent.	1	25,348	11,288	13,688	24,976	1.015
Chamber	2	35,802	15,199	21,254	36,453	0.982
Matching	1	37,951	16,732	17,168	33,900	1.119
Chiral	1	29,599	8,831	23,171	32,002	0.925
Operations	3	200,891	58,215	100,379	158,594	1.267
Gluing	1	27,081	28,416	11,565	39,981	0.677
Rank contr.	1	15,530	6,937	10,555	17,492	0.888
Locally proj.	2	26,490	25,498	11,966	37,464	0.707
Mathieu	1	13,736	5,516	35,862	41,378	0.332
Rank augm.	1	33,604	13,904	14,096	28,000	1.200
Mixes	1	57,753	12,615	31,077	43,692	1.322

Scope. The active catalogue contains 106 accessible originals: 43 admitted principal-result recoveries with measurements and 63 awaiting complete recovery. The 55 unavailable originals from the historical catalogue of 161 remain archived outside the analysis. No singleton is excluded. An admission covers the specified principal results, corrections and retained inputs; it does not establish every auxiliary claim or answer every conjecture.

Representations. B and C include complete authored-text renderings where original PDFs are unavailable. F, Mixes, rank contraction, rank augmentation and the Mathieu census use raw untrimmed PDFs. Other topics retain their declared calibrated and selected text. No pooled factor is formed across these different representations.

The complete common theory and every application. The dark bar is the sum of the original manuscript costs. Blue segments are complete common papers; green segments are complete applications. All companions and complete written tables remain charged.



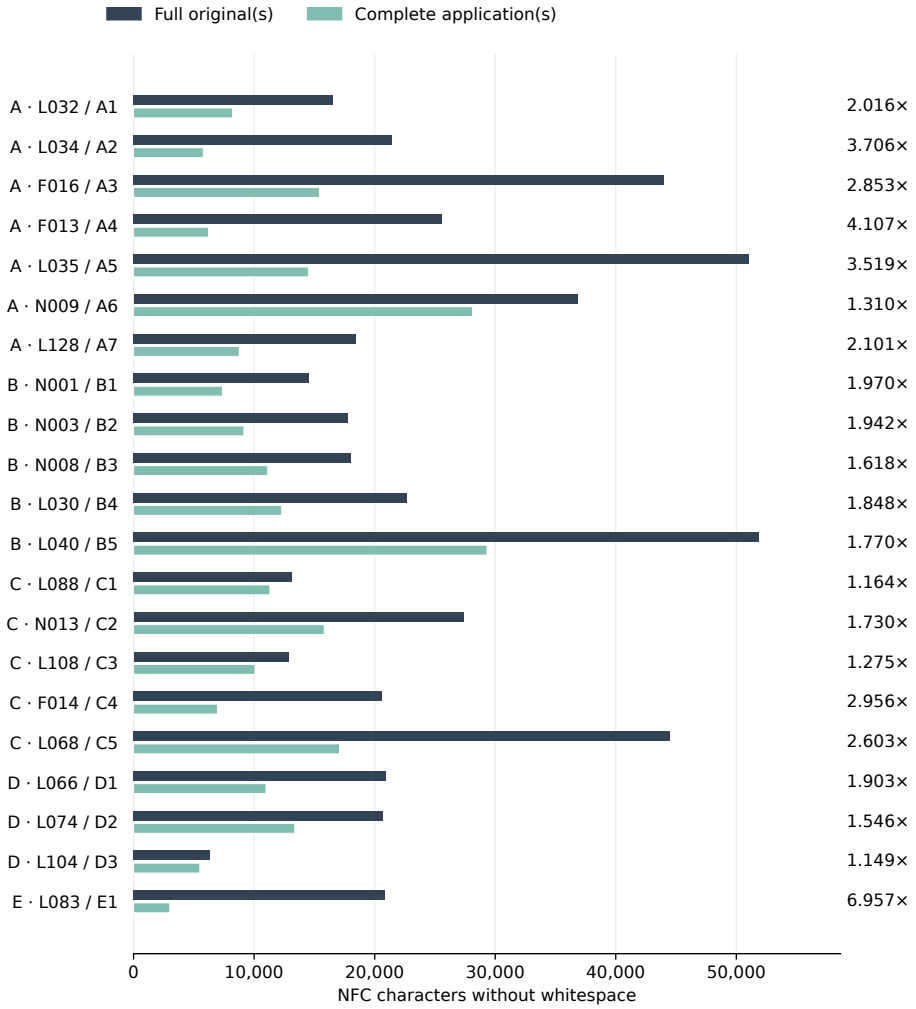
F now comprises two originals. Its entire seven-page common theory costs 15,876 characters and its five- and six-page applications cost 11,504 and 15,894. Thus 69,825/43,274 = 1.613555 .... The eighteen other topics retain their previous written costs. This is a measurement of the present proofs' written length, not of minimum proof complexity or originality.

Each complete application after acquisition. The entire common theory and all required dependencies have already been acquired. Optional validation is not a mathematical prerequisite. Every application, companion and reference is still charged.

Topic	Original	Application	Source cost	Application cost	Conditional
A	L032	A1	16,558	8,212	2.016
A	L034	A2	21,433	5,784	3.706
A	F016	A3	43,983	15,419	2.853
A	F013	A4	25,562	6,224	4.107
A	L035	A5	51,019	14,498	3.519
A	N009	A6	36,824	28,114	1.310
A	L128	A7	18,424	8,768	2.101
B	N001	B1	14,519	7,371	1.970
B	N003	B2	17,780	9,155	1.942
B	N008	B3	18,009	11,132	1.618
B	L030	B4	22,704	12,286	1.848
B	L040	B5	51,894	29,321	1.770
C	L088	C1	13,166	11,311	1.164
C	N013	C2	27,361	15,815	1.730
C	L108	C3	12,849	10,075	1.275
C	F014	C4	20,577	6,962	2.956
C	L068	C5	44,480	17,085	2.603
D	L066	D1	20,916	10,989	1.903
D	L074	D2	20,661	13,367	1.546
D	L104	D3	6,317	5,499	1.149
E	L083	E1	20,885	3,002	6.957
E	L079	E2	20,908	1,855	11.271
E	L021	E3	29,582	14,656	2.018
F	L014	F1	25,574	11,504	2.223
F	L007	F2	44,251	15,894	2.784
G	L009	G1	49,891	23,544	2.119
G	L092	G2	23,308	61,663	0.378
Affine	L012	AF1	32,767	22,638	1.447
Represent.	F010	R1	25,348	13,688	1.852
Chamber	L008	H1	21,784	7,571	2.877
Chamber	N016	H2	14,018	13,683	1.024
Matching	F015	M1	37,951	17,168	2.211
Chiral	L028	CE1	29,599	23,171	1.277
Operations	N005	OP1	122,002	49,595	2.460
Operations	N006	DG2	56,001	38,479	1.455
Operations	N007	DG1	22,888	12,305	1.860
Gluing	L010	GL1	27,081	11,565	2.342
Rank contr.	L037	RC1	15,530	10,555	1.471
Locally proj.	L084	LP1	2,071	4,857	0.426
Locally proj.	L095	LP2	24,419	7,109	3.435
Mathieu	L073	MC1	13,736	35,862	0.383
Rank augm.	L020	RA1	33,604	14,096	2.384
Mixes	F005	MX1	57,753	31,077	1.858

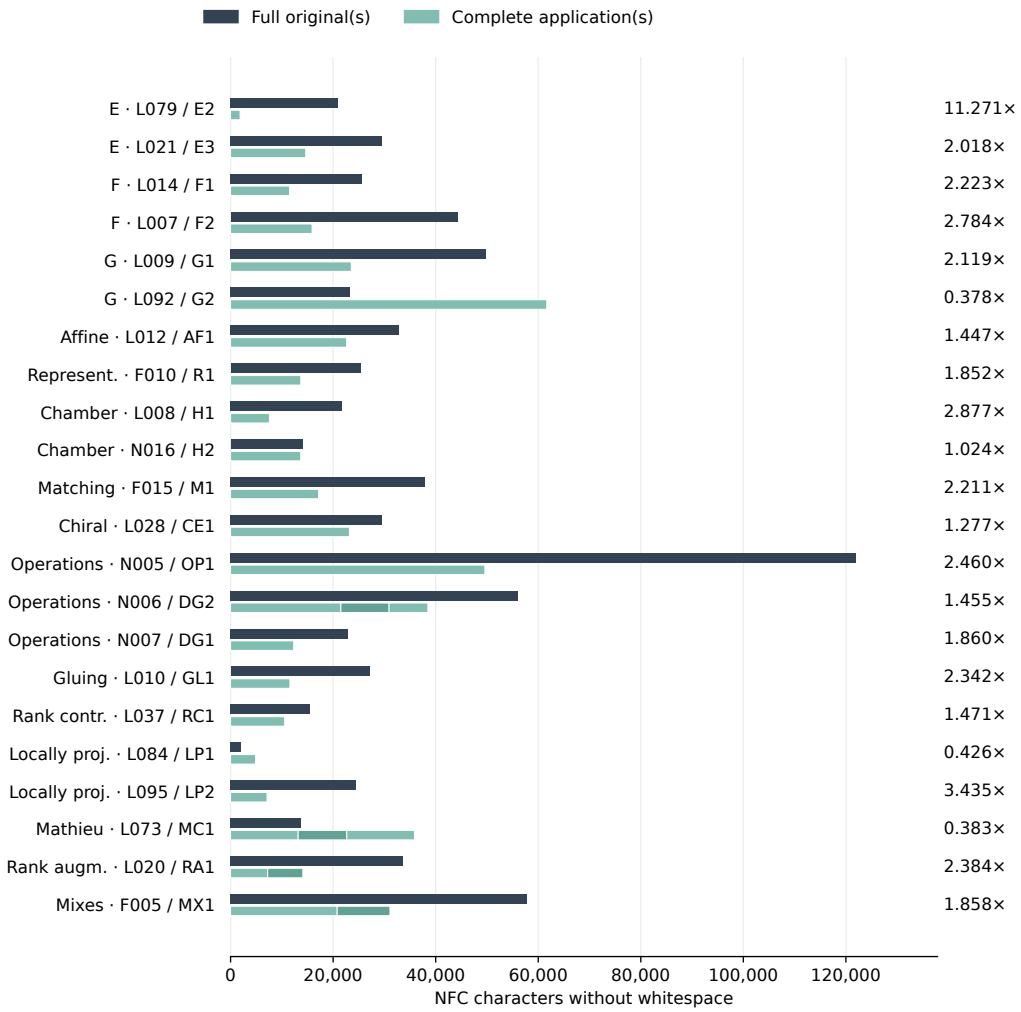
DG2 includes all N006 companions. MC1 includes all 24 Mathieu application pages. RA1 includes its application and counterproof; MX1 includes both local papers. F1 and F2 use the same whole common theory. In F, all nine programs are optional validation and none is a premise of the uniform proofs.

Original and application costs: first twenty-one originals. Dark bars show whole original manuscripts and green bars show whole applications. The bars share one scale within each chart. The factors assume prior acquisition of the complete common material.



The numerical input retains the exact edition, source qualifications and all alternative counting coordinates for each topic. F's primary coordinate now counts raw page furniture consistently; the old selected L014 cost of 25,115 is not substituted for its current raw cost of 25,574.

Original and application costs: remaining twenty-two originals. The same complete-document convention applies. A large conditional factor does not mean that the shared theory was free to establish.



The locally projective source L084 is the complete erratum; L095 uses the available fifteen-page arXiv edition. The Mathieu application contains the corrected 81/431-class census and every literal source-row disposition. L020's false third case has a corrected criterion and complete counterproof. The Mixes paper's three final problems remain unanswered. These qualifications remain part of the scientific admissions.

Program and data inventory. Validation must be excluded on both sides or included on both sides. The earlier software-inclusive fractions added only our programs and are withdrawn as compression measures. An inclusive comparison requires

$$\frac{S + c(\text{original validation})}{U + A + c(\text{our validation})}.$$

Original software costs are not yet established under a common inclusion rule, so this ratio is unavailable. The following replacement costs remain visible as inventory. Printed and raw programs are alternatives.

Topic	Printed programs	Raw programs	Mandatory data
A	54,210	52,807	–
B	13,170	12,330	–
C	20,933	19,937	–
D	27,070	26,185	–
E	7,565	–	–
F	22,466	21,691	–
G	53,018	50,184	–
Affine	17,250	16,682	–
Represent.	9,025	8,310	–
Chamber	7,677	6,401	–
Matching	4,998	4,408	–
Chiral	11,940	11,090	–
Operations	–	403,848	–
Gluing	13,371	12,768	–
Rank contr.	2,925	2,501	–
Locally proj.	22,695	22,104	28,574
Mathieu	31,244	30,489	48,686
Rank augm.	10,866	9,998	–
Mixes	16,025	14,608	–

E additionally retains 5,173 characters of optional printed validation, separate from its 7,565-character required program. F's printed and raw entries both represent all nine optional validation files; they are not additional to each other. Operations' whole raw inventory includes both required decoding programs. Its separately inventoried external data uses byte units, which are not added to these character costs.

Mathematical role. A program, certificate or data file needed to establish a theorem remains required evidence. A manuscript-length factor does not measure the complete cost of an executable proof when such premises lie outside the manuscript. Missing source software, full external proofs and graphical information retain unknown costs. Partial Lean certificates do not establish full formalization of the source theorems.

Embedded listings. Separate validation is excluded from both manuscript operands. F's two original papers and three replacement papers were checked for embedded validation listings and contain none. This specific check has not yet been completed for all older topics, so their unchanged complete-manuscript figures are not universally described as program-free comparisons.

Evidence. Exact integers, rational comparisons, cost inventories and document boundaries are supplied in the accompanying JSON and CSV files. This overview adds no proof, conjecture answer or formal declaration. Distinct review of the consolidation is required before publication.